

Hysol MB 50

Section 1. Identification

Product name	Hysol MB 50
Product code	465450-AU22
SDS no.	465450
Use of the substance/mixture	Metalworking fluid - soluble. For specific application advice see appropriate Technical Data Sheet or consult our company representative.
Product type	Liquid.
Supplier	Castrol New Zealand Limited 73 Remuera Road Newmarket Auckland, New Zealand www.castrol.com/nz Technical Helpline 0800 10 40 60
Emergency telephone number	0800 243643 (0800 CHEMHELP) (NZ use only)
New Zealand National Poisons Centre	0800 764 766 National Poison Centre

Section 2. Hazards identification

HSNO Classification	SKIN IRRITATION - Category 2 SERIOUS EYE DAMAGE - Category 1 SKIN SENSITISATION - Category 1
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This material is classified as hazardous according to criteria in the Hazardous Substances (Hazard Classification) Notice 2020.

This material is not classified as DANGEROUS GOODS according to criteria in New Zealand Standard 5433:2012 Transport of Dangerous Goods on Land.

Routes of entry	Dermal contact. Eye contact. Inhalation.
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GHS label elements

Signal word	Danger
Hazard statements	Causes skin irritation. May cause an allergic skin reaction. Causes serious eye damage.

Precautionary statements

Prevention	Wear protective gloves. Wear eye or face protection. Avoid breathing vapour. Wash hands thoroughly after handling. Contaminated work clothing should not be allowed out of the workplace.
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Response	Take off contaminated clothing and wash it before reuse. IF ON SKIN: Wash with plenty of soap and water. If skin irritation or rash occurs: Get medical attention. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER or physician.
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Storage	Not applicable.
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Disposal	Dispose of contents and container in accordance with all local, regional, national and international regulations.
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Symbol

Section 2. Hazards identification

Other hazards which do not result in classification

Defatting to the skin.

Section 3. Composition/information on ingredients

Substance/mixture

Mixture

Highly refined mineral oil, emulsifiers and additives.

Ingredient name	% (w/w)	CAS number
Distillates (petroleum), hydrotreated heavy naphthenic	≥30 - ≤60	64742-52-5
Amine neutralised carboxylic acids	≥10 - ≤30	-
Amine neutralised phosphoric acid esters	≤5	-
2-Amino-2-methylpropanol	≤3.4	124-68-5
Amine neutralised carboxylic acids	<3	-
2,2',2''-nitrilotriethanol	≤3	102-71-6
boric acid	≤1	10043-35-3
3-Iodo-2-propynyl butylcarbamate	≤0.21	55406-53-6

There are no additional ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment and hence require reporting in this section.

Occupational exposure limits, if available, are listed in Section 8.

Section 4. First aid measures

Description of necessary first aid measures

Inhalation

If inhaled, remove to fresh air. In case of inhalation of decomposition products in a fire, symptoms may be delayed. The exposed person may need to be kept under medical surveillance for 48 hours. Get medical attention immediately.

Ingestion

Do not induce vomiting unless directed to do so by medical personnel. Never give anything by mouth to an unconscious person. If unconscious, place in recovery position and get medical attention immediately. Wash out mouth with water if person is conscious. Get medical attention immediately. Chemical burns must be treated promptly by a physician.

Skin contact

Get medical attention immediately. In case of contact, immediately flush skin with plenty of water for at least 15 minutes while removing contaminated clothing and shoes. Wash skin thoroughly with soap and water or use recognised skin cleanser. Wash clothing before reuse. Wash contaminated clothing thoroughly with water before removing it, or wear gloves. Clean shoes thoroughly before reuse. In the event of any complaints or symptoms, avoid further exposure. Chemical burns must be treated promptly by a physician.

Eye contact

In case of contact, immediately flush eyes with plenty of water for at least 15 minutes. Eyelids should be held away from the eyeball to ensure thorough rinsing. Check for and remove any contact lenses. Get medical attention immediately. Chemical burns must be treated promptly by a physician.

Indication of immediate medical attention and special treatment needed, if necessary

Notes to physician

Treatment should in general be symptomatic and directed to relieving any effects. In case of inhalation of decomposition products in a fire, symptoms may be delayed. The exposed person may need to be kept under medical surveillance for 48 hours.

Protection of first-aiders

No action shall be taken involving any personal risk or without suitable training. If it is suspected that fumes are still present, the rescuer should wear an appropriate mask or self-contained breathing apparatus. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Wash contaminated clothing thoroughly with water before removing it, or wear gloves.

Section 5. Firefighting measures

Extinguishing media

Suitable	In case of fire, use water fog, alcohol resistant foam, dry chemical or carbon dioxide extinguisher or spray.
Not suitable	Do not use water jet.
Specific hazards arising from the chemical	In a fire or if heated, a pressure increase will occur and the container may burst.
Hazardous combustion products	Combustion products may include the following: phosphorus oxides carbon oxides (CO, CO ₂) (carbon monoxide, carbon dioxide) nitrogen oxides (NO, NO ₂ etc.)
Hazchem code	Not available.
Special precautions for fire-fighters	No action shall be taken involving any personal risk or without suitable training. Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire.
Special protective equipment for fire-fighters	Fire-fighters should wear positive pressure self-contained breathing apparatus (SCBA) and full turnout gear.

Section 6. Accidental release measures

Personal precautions, protective equipment and emergency procedures

For non-emergency personnel	Entry into a confined space or poorly ventilated area contaminated with vapour, mist or fume is extremely hazardous without the correct respiratory protective equipment and a safe system of work. Wear self-contained breathing apparatus. Wear a suitable chemical protective suit. Chemical resistant boots. See also the information in "For non-emergency personnel".
For emergency responders	Entry into a confined space or poorly ventilated area contaminated with vapour, mist or fume is extremely hazardous without the correct respiratory protective equipment and a safe system of work. Wear self-contained breathing apparatus. Wear a suitable chemical protective suit. Chemical resistant boots. See also the information in "For non-emergency personnel".
Environmental precautions	Avoid dispersal of spilt material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air).

Methods and material for containment and cleaning up

Small spill	Stop leak if without risk. Move containers from spill area. Absorb with an inert material and place in an appropriate waste disposal container. Dispose of via a licensed waste disposal contractor.
Large spill	Stop leak if without risk. Move containers from spill area. Approach the release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Contain and collect spillage with non-combustible, absorbent material e.g. sand, earth, vermiculite or diatomaceous earth and place in container for disposal according to local regulations. Contaminated absorbent material may pose the same hazard as the spilt product. Dispose of via a licensed waste disposal contractor.

Section 7. Handling and storage

Precautions for safe handling	Put on appropriate personal protective equipment (see Section 8). Persons with a history of skin sensitization problems should not be employed in any process in which this product is used. Do not get in eyes or on skin or clothing. Do not breathe vapour or mist. Do not ingest. If during normal use the material presents a respiratory hazard, use only with adequate ventilation or wear appropriate respirator. Keep in the original container or an approved alternative made from a compatible material, kept tightly closed when not in use. Empty containers retain product residue and can be hazardous. Do not reuse container. Avoid contact of spilt material and runoff with soil and surface waterways. Avoid prolonged or repeated contact with skin. During metal working, solid particles from workpieces or tools will contaminate the fluid and may cause abrasions of the skin. Where such abrasions result in a penetration of the skin, first aid treatment should be applied as soon as reasonably possible. The presence of certain metals in the workpiece or tool, such as chromium, cobalt and nickel, can contaminate the metalworking fluid and as a
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Section 7. Handling and storage

Conditions for safe storage, including any incompatibilities

result may induce allergic skin reactions. Evaporation of water from soluble cutting fluids during use may lead to an increase in concentration which may result in the development of skin conditions due to irritation and defatting. It is important to monitor fluid strength on a regular basis with a refractometer and maintain it at the recommended concentration. Lubricants from other sources and other contaminants should be minimised. Swarf and other debris should be removed. To maintain optimum performance and minimise bacterial spoilage, machine tool coolant systems should be cleaned on a regular basis.

Store in accordance with local regulations. Store in original container protected from direct sunlight in a dry, cool and well-ventilated area, away from incompatible materials (see Section 10) and food and drink. Store locked up. Keep container tightly closed and sealed until ready for use. Store and use only in equipment/containers designed for use with this product. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Do not store in unlabelled containers. Use appropriate containment to avoid environmental contamination. DO NOT ADD NITRITES TO THIS FLUID.

Section 8. Exposure controls/personal protection

Control parameters

Occupational exposure limits

Ingredient name	Exposure limits
Distillates (petroleum), hydrotreated heavy naphthenic	NZ HSWA 2015 - GRWM 2016 (New Zealand). WES-TWA: 5 mg/m ³ 8 hours. Issued/Revised: 6/2016 Form: Mist WES-STEL: 10 mg/m ³ 15 minutes. Issued/Revised: 9/2010 Form: Mist
2,2',2''-nitrilotriethanol	NZ HSWA 2015 - GRWM 2016 (New Zealand). WES-TWA: 5 mg/m ³ 8 hours. Issued/Revised: 1/1994
Boric acid	ACGIH TLV (United States). TWA: 2 mg/m ³ 8 hours. Issued/Revised: 1/2005 Form: Inhalable fraction STEL: 6 mg/m ³ 15 minutes. Issued/Revised: 1/2005 Form: Inhalable fraction

Biological exposure indices

No exposure indices known.

Recommended monitoring procedures

If this product contains ingredients with exposure limits, personal, workplace atmosphere or biological monitoring may be required to determine the effectiveness of the ventilation or other control measures and/or the necessity to use respiratory protective equipment. Reference should be made to appropriate monitoring standards. Reference to national guidance documents for methods for the determination of hazardous substances will also be required.

Appropriate engineering controls

All activities involving chemicals should be assessed for their risks to health, to ensure exposures are adequately controlled. Personal protective equipment should only be considered after other forms of control measures (e.g. engineering controls) have been suitably evaluated. Personal protective equipment should conform to appropriate standards, be suitable for use, be kept in good condition and properly maintained.

Your supplier of personal protective equipment should be consulted for advice on selection and appropriate standards. For further information contact your national organisation for standards.

Provide exhaust ventilation or other engineering controls to keep the relevant airborne concentrations below their respective occupational exposure limits.

The final choice of protective equipment will depend upon a risk assessment. It is important to ensure that all items of personal protective equipment are compatible.

Section 8. Exposure controls/personal protection

Environmental exposure controls

Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

Individual protection measures

Hygiene measures

Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Appropriate techniques should be used to remove potentially contaminated clothing. Contaminated work clothing should not be allowed out of the workplace. Wash contaminated clothing before reusing. Ensure that eyewash stations and safety showers are close to the workstation location.

Eye protection

Undiluted fluid: Chemical goggles.
Diluted fluid: Safety glasses with side shields.

Hand protection

Wear suitable gloves. Undiluted fluid: Wear chemical resistant gloves. Recommended: nitrile gloves.
Diluted fluid: Wear protective gloves if prolonged or repeated contact is likely. Recommended: nitrile gloves. The correct choice of protective gloves depends upon the chemicals being handled, the conditions of work and use, and the condition of the gloves (even the best chemically resistant glove will break down after repeated chemical exposures). Most gloves provide only a short time of protection before they must be discarded and replaced. Because specific work environments and material handling practices vary, safety procedures should be developed for each intended application. Gloves should therefore be chosen in consultation with the supplier/manufacturer and with a full assessment of the working conditions.

Skin protection

Appropriate footwear and any additional skin protection measures should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.

Respiratory protection

In case of insufficient ventilation, wear suitable respiratory equipment. For protection against metal working fluids, respiratory protection that is classified as "resistant to oil" (class R) or oil proof (class P) should be selected where appropriate. Depending on the level of airborne contaminants, an air-purifying, half-mask respirator (with HEPA filter) including disposable (P- or R-series) (for oil mists less than 50mg/m³), or any powered, air-purifying respirator equipped with hood or helmet and HEPA filter (for oil mists less than 125 mg/m³). Where organic vapours are a potential hazard during metalworking operations, a combination particulate and organic vapour filter may be necessary. The correct choice of respiratory protection depends upon the chemicals being handled, the conditions of work and use, and the condition of the respiratory equipment. Safety procedures should be developed for each intended application. Respiratory protection equipment should therefore be chosen in consultation with the supplier/manufacturer and with a full assessment of the working conditions.

Section 9. Physical and chemical properties

The conditions of measurement of all properties are at standard temperature and pressure unless otherwise indicated.

Appearance

Physical state

Liquid.

Colour

Yellow. [Light]

Odour

Not available.

pH

9.4 to 9.9 [Conc. (% w/w): 5%]

Melting point/freezing point

Not available.

Boiling point, initial boiling point, and boiling range

Not available.

Drop Point

Not available.

Flash point

Closed cup: >100°C (>212°F) [Estimated. Water content interferes with flash point determination.]

Auto-ignition temperature

Section 9. Physical and chemical properties

Ingredient name	°C	°F	Method
2-Amino-2-methylpropanol	438	820.4	ASTM D 2161
2,2',2''-nitritotriethanol	324	615.2	

Vapour pressure

Ingredient name	Vapour Pressure at 20°C			Vapour pressure at 50°C		
	mm Hg	kPa	Method	mm Hg	kPa	Method
Distillates (petroleum), hydrotreated heavy naphthenic	<0.08	<0.011	ASTM D 5191			
Water	23.8	3.2	ASTM E 1194			
2-Amino-2-methylpropanol	0.34	0.045				
2,2',2''-nitritotriethanol	<0.01	<0.0013				

Relative vapour density

Not available.

Density

<1000 kg/m³ (<1 g/cm³) at 15°C

Solubility(ies)

Not available.

Particle characteristics

Median particle size

Not applicable.

Section 10. Stability and reactivity

Chemical stability

The product is stable.

Possibility of hazardous reactions

Under normal conditions of storage and use, hazardous reactions will not occur.
Under normal conditions of storage and use, hazardous polymerisation will not occur.

Conditions to avoid

Avoid excessive heat.

Incompatible materials

Reactive or incompatible with the following materials: oxidising materials.
Slightly reactive or incompatible with the following materials: acids.

Hazardous decomposition products

Under normal conditions of storage and use, hazardous decomposition products should not be produced.

Section 11. Toxicological information

Information on likely routes of exposure

Inhalation

May give off gas, vapour or dust that is very irritating or corrosive to the respiratory system. Exposure to decomposition products may cause a health hazard. Serious effects may be delayed following exposure.

Ingestion

May cause burns to mouth, throat and stomach.

Skin contact

Causes skin irritation. Defatting to the skin. May cause an allergic skin reaction.

Eye contact

Causes serious eye damage.

Symptoms related to the physical, chemical and toxicological characteristics

Inhalation

No specific data.

Ingestion

Adverse symptoms may include the following:
stomach pains

Skin contact

Adverse symptoms may include the following:
irritation
redness
dryness
cracking

Eye contact

Adverse symptoms may include the following:
pain
watering
redness

Potential chronic health effects

Section 11. Toxicological information

General	No known significant effects or critical hazards.
Inhalation	Not applicable.
Ingestion	Not applicable.
Skin contact	Not applicable.
Eye contact	Not applicable.
Carcinogenicity	No known significant effects or critical hazards.
Mutagenicity	No known significant effects or critical hazards.
Teratogenicity	No known significant effects or critical hazards.
Developmental effects	No known significant effects or critical hazards.
Fertility effects	No known significant effects or critical hazards.

Section 12. Ecological information

Ecotoxicity No known significant effects or critical hazards.

Persistence and degradability

Expected to be biodegradable.

Bioaccumulative potential

Not available.

Product/ingredient name	LogP _{ow}	BCF	Potential
2-Amino-2-methylpropanol	-0.63	-	low
2,2',2''-nitrilotriethanol	-1	-	low
Boric acid	-1.09	-	low

Mobility in soil

Mobility Liquid. Soluble in water.

Soil/water partition coefficient (K_{oc}) Not available.

Section 13. Disposal considerations

Disposal methods

The generation of waste should be avoided or minimised wherever possible. Significant quantities of waste product residues should not be disposed of via the foul sewer but processed in a suitable effluent treatment plant. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible. This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Avoid dispersal of spilt material and runoff and contact with soil, waterways, drains and sewers. Diluted Fluid The spent diluted fluid comprises a relatively stable emulsion. Dispose of via an authorised person/ licensed waste disposal contractor or by other suitable waste treatment techniques (e.g. emulsion splitting, coagulation and filtration) approved by the local authority. Spent fluid should never be disposed of down the drain. The aqueous phase should not be discharged into sewage systems unless provided for by local regulations; the non-aqueous phase should be disposed of as undiluted fluid. Note that separated aqueous solutions or effluents may contain metal salts as well as traces of oil and must be checked for conformity in these respects against consents given by the authorities before disposal. Further treatment may be required.

Section 14. Transport information

Regulatory information	UN number	Proper shipping name	Classes	PG*	Label	Additional information
New Zealand Class	Not regulated.	-	-	-		-
ADG Class	Not regulated.	-	-	-		-
IATA Class	Not regulated.	-	-	-		-
IMDG Class	Not regulated.	-	-	-		-

PG* : Packing group

Section 15. Regulatory information

New Zealand Regulatory Information

HSNO Approval Number	HSR002606
HSNO Group Standard	Lubricants, Lubricant Additives, Coolants and Anti-freeze Agents (Subsidiary Hazard) Group Standard 2020
HSNO Classification	SKIN IRRITATION - Category 2 SERIOUS EYE DAMAGE - Category 1 SKIN SENSITISATION - Category 1

Regulation according to other foreign laws

REACH Status	For the REACH status of this product please consult your company contact, as identified in Section 1.
United States inventory (TSCA 8b)	Not determined.
Australia inventory (AIC)	All components are listed or exempted.
Canada inventory status	Not determined.
China inventory (IECSC)	Not determined.
Japan inventory (CSCL)	At least one component is not listed.
Korea inventory (KECI)	Not determined.
Philippines inventory (PICCS)	At least one component is not listed.
Taiwan Chemical Substances Inventory (TCSI)	Not determined.

Section 16. Other information

History

Date of issue/Date of revision	16 August 2023
Date of previous issue	29 July 2022.
Version	3.01
Prepared by	Not available.
Key to abbreviations	Varies = may contain one or more of the following 64741-88-4, 64741-89-5, 64741-95-3, 64741-96-4, 64742-01-4, 64742-44-5, 64742-45-6, 64742-52-5, 64742-53-6, 64742-54-7, 64742-55-8, 64742-56-9, 64742-57-0, 64742-58-1, 64742-62-7, 64742-63-8, 64742-65-0, 64742-70-7, 72623-85-9, 72623-86-0, 72623-87-1

Notice to reader

Indicates information that has changed from previously issued version.

Section 16. Other information

All reasonably practicable steps have been taken to ensure this data sheet and the health, safety and environmental information contained in it is accurate as of the date specified below. No warranty or representation, express or implied is made as to the accuracy or completeness of the data and information in this data sheet.

The data and advice given apply when the product is sold for the stated application or applications. You should not use the product other than for the stated application or applications without seeking advice from BP Group.

It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. The BP Group shall not be responsible for any damage or injury resulting from use, other than the stated product use of the material, from any failure to adhere to recommendations, or from any hazards inherent in the nature of the material. Purchasers of the product for supply to a third party for use at work, have a duty to take all necessary steps to ensure that any person handling or using the product is provided with the information in this sheet. Employers have a duty to tell employees and others who may be affected of any hazards described in this sheet and of any precautions that should be taken. You can contact the BP Group to ensure that this document is the most current available. Alteration of this document is strictly prohibited.

Castrol® Hysol® MB 50

Your answer to the challenges of modern manufacturing



Multi-applications, multiple metals and harsh operating conditions make it difficult to keep metal-working fluids stable. Tramp oil, swarf and waste can leach into the system and promote bacterial growth, reduce the life of the fluid, compromise performance, and ultimately cause the system to fail. Combating this requires constant maintenance intervention that add complexity and cost to your operations. Castrol® Hysol® MB 50 makes a difference in the way you work, saving you time and money.

Castrol® Hysol® MB 50 is formulated to help you:

- Extend fluid life through a robust pH buffer package
- Reduce maintenance burden and machine downtime
- Minimize metalworking fluid usage, limit waste and reduce overall chemical spend
- Increase tramp oil rejection for easier removal, cleaner running systems and increased part cleanliness

Applicable sectors

Aerospace, Automotive, Agriculture, Construction, Engine Manufacturing, Heavy Industry, Job Shops, Machine Manufacturing, Medical, Primary and Secondary Metals, Precision Machining

Typical characteristics

Concentrate

Mineral Oil	High
Ester	Yes
Boron	Yes
Amine	Yes
Chlorides	No
Biocide	No

Multi-purpose product capabilities

When your business performs a variety of operations over a range of different metals, using a single coolant often seems unachievable and with compromises between quality, practicality and the demand of your operations.

Developed to solve multiple challenges, Hysol® MB 50 can take you from a position of compromise to one of strength. It gives you the advantages of a single multi-metal, multi-purpose metalworking fluid with all of the expertise of Castrol® engineering behind it.

Emulsion Properties

Refractometer Factor	1.1
Concentration (%)	5 - 10
Appearance	Translucent Milky
pH – 5% in DI Water	9.3 - 9.8
Cast Iron Chip Break Point	3%
Water Tolerance, gpg	5-35

Address your machining
challenges with

Castrol[®] Hysol[®] MB 50



Challenge: Coolant life

Castrol[®] Hysol[®] MB 50 is formulated with a resilient additive technology that resists the negative effects of bacterial growth and contamination. Your metalworking systems maintain performance, so you can rely on them to work harder for longer.

Increase productivity:

- Less downtime for cleaning
- Reduce frequency of coolant top-ups

Lower costs:

- Reduce maintenance for cleaning contaminated systems
- Coolant lasts longer with lower waste disposal

Safer working environment:

- Bacteria-resistant technology reduces the need for tank side biocides



Challenge: Multiple coolants, multi-metal machining

Castrol[®] Hysol[®] MB 50 delivers consistent performance across multiple machining processes and metals. You can run your shop on a single coolant, simplify operations and make it easier to meet business demands.

Increase productivity:

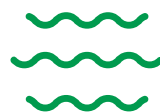
- Deliver consistently high quality surface finishes balanced with efficient production
- Eliminate risk of errors during top-ups with different coolants in the shop

Lower costs:

- Less downtime and manpower for coolant changeout
- Lowers inventory complexity by reducing the amount of metal-working fluids that are needed

Safer working environment:

- Reduce use of additives such as biocides and pH boosters



Challenge: Water hardness tolerance

Castrol[®] Hysol[®] MB 50 is designed to operate across a range of water hardness (100 – 600 ppm CaCO₃). It resists foaming and provides superior emulsion stability, reducing maintenance needs and simplifying operations.

Increase productivity:

- Reduce downtime for cleaning, coolant refresh and filter changes
- Reduce stoppages due to foam

Lower costs:

- Extend tool life, reduce need for maintenance and spares
- Reduce water treatment management

Safer working environment:

- Cleaner workplace and less chemical handling

SAFETY DATA SHEET

HOCUT™ 795-H

SDS according to the U.S. OSHA Hazard Communication Standard (29 CFR 1910.1200), Revision 2012

Section 1. Identification

Product code : 202374-01
Product name : HOCUT™ 795-H
Other means of identification : Not available.

Relevant identified uses of the substance or mixture and uses advised against

Relevant uses : Metalworking fluid
Uses advised against : Any other purpose.

Supplier : Quaker Houghton PA, Inc.
901 E. Hector Street
Conshohocken, PA 19428 USA
T: 610-832-4000

Wallover Oil Company
21845 Drake Road
Strongsville, OH 44149 USA
www.wallover.com
T: (440) 238-9250

ProductStewardship@quakerhoughton.com
www.quakerhoughton.com

Emergency telephone number (with hours of operation) : CHEMTREC US/Canada: 1-800-424-9300 or 1-703-527-3887 (24 hours)

Section 2. Hazards identification

OSHA/HCS status : This material is considered hazardous by the OSHA Hazard Communication Standard (29 CFR 1910.1200).
Classification of the substance or mixture : SKIN IRRITATION - Category 2
EYE IRRITATION - Category 2A
SKIN SENSITIZATION - Category 1

GHS label elements

Hazard pictograms :



Signal word : Warning
Hazard statements : Causes skin irritation.
May cause an allergic skin reaction.
Causes serious eye irritation.

Section 2. Hazards identification

Precautionary statements

Prevention	: Wear eye or face protection. Avoid breathing vapor. Wash thoroughly after handling.
Response	: Take off contaminated clothing and wash it before reuse. Wash contaminated clothing before reuse. IF ON SKIN: Wash with plenty of water. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice or attention.
Storage	: Not applicable.
Disposal	: Dispose of contents and container in accordance with all local, regional, national and international regulations.
Hazards not otherwise classified	: None known.

Section 3. Composition/information on ingredients

Substance/mixture : Mixture

Ingredient name	%	CAS number
Mineral oil	≥50 - ≤75	**
1-aminopropan-2-ol	≤10	78-96-6
Amine neutralized carboxylic Acid	≤5	-
Amine neutralized carboxylic Acid	≤5	-
Amine neutralized dicarboxylic Acid	≤3	-
Alcohols, C12-15, ethoxylated	≤3	68131-39-5
Amine neutralized carboxylic Acid	≤3	-
Amine neutralized carboxylic Acid	≤2.3	-
1,2-benzisothiazol-3(2H)-one	≤0.1	2634-33-5

**** May contain** : 64742-52-5,64742-53-6

The mineral oils in the product contain < 3% DMSO extract (IP 346).

The exact percentage (concentration) of composition has been withheld as a trade secret

Section 4. First aid measures

Description of necessary first aid measures

General advice	: Get medical attention. If medical advice is needed, have product container or label at hand. Use personal protective equipment as required. Remove contaminated clothing and wash it before reuse. Wash skin surfaces thoroughly after contact.
Inhalation	: Move affected person to fresh air. If not breathing, if breathing is irregular or if respiratory arrest occurs, provide artificial respiration or oxygen by trained personnel. Get medical attention.
Skin contact	: Wash with plenty of soap and water. Remove contaminated clothing and wash it before reuse. Get medical attention if symptoms occur.
Eye contact	: Flush with plenty of water for at least 15 minutes, occasionally lifting the upper and lower eyelids. Remove contact lenses, if present and easy to do. Get medical attention if symptoms occur.
Ingestion	: Ingestion may cause gastrointestinal irritation and diarrhea. Do not induce vomiting unless directed to do so by medical personnel. Never give anything by mouth to an unconscious person.

Most important symptoms and effects, both acute and delayed

Inhalation	: Not expected under normal use.
Skin contact	: pain or irritation, redness, skin rash or hives
Eye contact	: pain or irritation, redness, watering
Ingestion	: Not expected under normal use.

Section 4. First aid measures

Indication of immediate medical attention and special treatment needed, if necessary

- Notes to physician** : In case of inhalation of decomposition products in a fire, symptoms may be delayed. The exposed person may need to be kept under medical surveillance for 48 hours.
- Specific treatments** : No specific treatment.
- Protection of first-aiders** : No action shall be taken involving any personal risk or without suitable training. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Wash contaminated clothing thoroughly with water before removing it, or wear gloves. Use personal protective equipment as required.

Section 5. Fire-fighting measures

Extinguishing media

- Suitable extinguishing media** : Use an extinguishing agent suitable for the surrounding fire. Use dry chemical, CO₂, water spray (fog) or foam.
- Unsuitable extinguishing media** : Do not use water jet.

- Specific hazards arising from the chemical** : In a fire or if heated, a pressure increase will occur and the container may burst.

- Hazardous thermal decomposition products** : In a fire, hazardous decomposition products may be produced. carbon oxides (CO, CO₂) nitrogen oxides

- Special protective actions for fire-fighters** : Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training.

- Special protective equipment for fire-fighters** : Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

Section 6. Accidental release measures

Personal precautions, protective equipment and emergency procedures

- For non-emergency personnel** : No action shall be taken involving any personal risk or without suitable training. Put on appropriate personal protective equipment (see Section 8). Keep unnecessary personnel away. Avoid breathing vapor or mist. Provide adequate ventilation.
- For emergency responders** : If specialized clothing is required to deal with the spillage, take note of any information in Section 8 on suitable and unsuitable materials. See also the information in "For non-emergency personnel". Evacuate area.

- Environmental precautions** : Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air). Do not allow any potentially contaminated water, including rain water, runoff from fire fighting or spills, to enter any waterway, sewer or drain.

Methods and materials for containment and cleaning up

- Small spill** : Stop leak if without risk. Move containers from spill area. Prevent entry into sewers, water courses, basements or confined areas. Absorb with an inert material and place in an appropriate waste disposal container. Dispose of via a licensed waste disposal contractor.
- Large spill** : Stop leak if without risk. Move containers from spill area. For large spills, dike spilled material or otherwise contain it to ensure runoff does not reach a waterway. Absorb with an inert material and place in an appropriate waste disposal container. Dispose of via a licensed waste disposal contractor.

Section 7. Handling and storage

Precautions for safe handling

- Protective measures** : Put on appropriate personal protective equipment (see Section 8). Do not get in eyes or on skin or clothing. Do not breathe vapor or mist. Do not ingest.
- Advice on general occupational hygiene** : Eating, drinking and smoking should be prohibited in areas where this material is handled, stored and processed. Workers should wash hands and face before eating, drinking and smoking. Remove contaminated clothing and protective equipment before entering eating areas. See also Section 8 for additional information on hygiene measures.
- Conditions for safe storage, including any incompatibilities** : Store in accordance with local regulations. Store in original container protected from direct sunlight in a dry, cool and well-ventilated area, away from incompatible materials (see Section 10) and food and drink. Do not store in unlabeled containers. Use appropriate containment to avoid environmental contamination.
- Storage temperature** : Store between the following temperatures: -5 to 50°C (23 to 122°F).

Section 8. Exposure controls/personal protection

Control parameters

Occupational exposure limits

Ingredient name	Exposure limits
Mineral oil	NIOSH REL (United States, 10/2016).
	TWA: 5 mg/m ³ 10 hours. Form: Mist
	STEL: 10 mg/m ³ 15 minutes. Form: Mist
	ACGIH TLV (United States).
	TWA: 5 mg/m ³ 8 hours.
	STEL: 10 mg/m ³ 15 minutes.
1-aminopropan-2-ol	None.
Amine neutralized carboxylic Acid	None.
Amine neutralized carboxylic Acid	None.
Amine neutralized dicarboxylic Acid	None.
Alcohols, C12-15, ethoxylated	None.
Amine neutralized carboxylic Acid	None.
Amine neutralized carboxylic Acid	None.
1,2-benzisothiazol-3(2H)-one	None.

- Appropriate engineering controls** : If user operations generate dust, fumes, gas, vapor or mist, use process enclosures, local exhaust ventilation or other engineering controls to keep worker exposure to airborne contaminants below any recommended or statutory limits.
- Environmental exposure controls** : Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

Individual protection measures

- Hygiene measures** : Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Eating, drinking and smoking should be prohibited in areas where this material is handled, stored and processed. Appropriate techniques should be used to remove potentially contaminated clothing. Wash contaminated clothing before reusing. Ensure that eyewash stations and safety showers are close to the workstation location. Keep equipment clean.
- Eye/face protection** : Safety eyewear complying with an approved standard should be used when a risk assessment indicates this is necessary to avoid exposure to liquid splashes, mists, gases or dusts. If contact is possible, the following protection should be worn, unless the assessment indicates a higher degree of protection: chemical splash goggles.

Section 8. Exposure controls/personal protection

Hand protection	: Chemical-resistant, impervious gloves complying with an approved standard should be worn at all times when handling chemical products if a risk assessment indicates this is necessary. Considering the parameters specified by the glove manufacturer, check during use that the gloves are still retaining their protective properties. It should be noted that the time to breakthrough for any glove material may be different for different glove manufacturers.
Other skin protection	: Personal protective equipment for the body should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product. Appropriate footwear and any additional skin protection measures should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.
Respiratory protection	: Based on the hazard and potential for exposure, select a respirator that meets the appropriate standard or certification. Respirators must be used according to a respiratory protection program to ensure proper fitting, training, and other important aspects of use.
Thermal hazards	: Not expected under normal use. Not relevant/applicable due to nature of the product.

Section 9. Physical and chemical properties

Appearance

Physical state	: Liquid.
Color	: Hazy
Odor	: Characteristic.
Odor threshold	: Not available.
pH	: 10.2 [Conc. (% w/w): 5%]
Melting point	: Not available.
Boiling point	: Not available.
Flash point	: Not available.
Evaporation rate	: Not available.
Flammability (solid, gas)	: Not available.
Lower and upper explosive (flammable) limits	: Not available.
Vapor pressure	: Not available.
Vapor density	: Not available.
Relative density	: 0.942
Solubility	: Emulsifies.
Solubility in water	: Not available.
Partition coefficient: n-octanol/water	: Not available.
Auto-ignition temperature	: Not available.
Decomposition temperature	: Not available.
Viscosity	: Kinematic (40°C (104°F)): >0.206 cm ² /s (>20.6 cSt)

VOC content

Product	: 142 g/l	ASTM E1868-10
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Section 10. Stability and reactivity

Reactivity	: No specific test data related to reactivity available for this product or its ingredients.
Chemical stability	: The product is stable.
Possibility of hazardous reactions	: Under normal conditions of storage and use, hazardous reactions will not occur.
Conditions to avoid	: No specific measures identified.
Incompatible materials	: Strong oxidizing materials. strong acids. strong alkalis
Hazardous decomposition products	: Under normal conditions of storage and use, hazardous decomposition products should not be produced.

Section 11. Toxicological information

Information on toxicological effects

Acute toxicity : Based on available data, the classification criteria are not met.

Acute toxicity estimates

Route	ATE value
Oral	4816.6 mg/kg
Dermal	10785.65 mg/kg

Numerical measures of toxicity

Product/ingredient name	Result	Species	Dose	Exposure
1-aminopropan-2-ol	LD50 Dermal	Rabbit	1851 mg/kg	-
	LD50 Oral	Rat	1715 mg/kg	-
Alcohols, C12-15, ethoxylated	LD50 Oral	Rat	2 g/kg	-
1,2-benzisothiazol-3(2H)-one	LC50 Inhalation Dusts and mists	Rat	0.0501 mg/l	4 hours
	LD50 Dermal	Rat	4115 mg/kg	-
	LD50 Oral	Rat	1020 mg/kg	-

Irritation/Corrosion : Causes severe eye irritation. Causes skin irritation.

Product/ingredient name	Result	Species	Score	Exposure	Observation
1-aminopropan-2-ol	Eyes - Severe irritant	Rabbit	-	24 hours 250 ug	-
	Eyes - Severe irritant	Rabbit	-	970 ug	-
	Skin - Mild irritant	Rabbit	-	24 hours 500 mg	-
1,2-benzisothiazol-3(2H)-one	Skin - Moderate irritant	Rabbit	-	485 mg	-
	Skin - Mild irritant	Human	-	48 hours 5 %	-

Sensitization : May cause sensitization by skin contact.

Mutagenicity : Based on available data, the classification criteria are not met.

Carcinogenicity : Based on available data, the classification criteria are not met.

Reproductive toxicity : Based on available data, the classification criteria are not met.

Specific target organ toxicity (single exposure) : Based on available data, the classification criteria are not met.

Specific target organ toxicity (repeated exposure) : Based on available data, the classification criteria are not met.

Aspiration hazard : Based on available data, the classification criteria are not met.

Other information : None identified.

Information on the likely routes of exposure

Section 11. Toxicological information

Inhalation	: No known significant effects or critical hazards.
Skin contact	: Causes skin irritation. May cause sensitization by skin contact.
Eye contact	: Causes serious eye irritation.
Ingestion	: No known significant effects or critical hazards.

Delayed and immediate effects and also chronic effects from short and long term exposure

None identified.

Symptoms related to the physical, chemical and toxicological characteristics

Inhalation	: Not expected under normal use.
Skin contact	: pain or irritation, redness, skin rash or hives
Eye contact	: pain or irritation, redness, watering
Ingestion	: Not expected under normal use.

Section 12. Ecological information

This material is toxic to aquatic life. This material is harmful to aquatic life with long lasting effects.

Toxicity

Product/ingredient name	Result	Species	Exposure
1-aminopropan-2-ol	Acute EC50 32.7 mg/l	Algae - Scenedesmus subspicatus	72 hours
Alcohols, C12-15, ethoxylated	Acute EC50 108.82 mg/l	Daphnia - Daphnia magna	48 hours
	Acute LC50 210 mg/l Fresh water	Fish - Carassius auratus	96 hours
	Acute EC50 0.7 mg/l Fresh water	Algae - Pseudokirchneriella subcapitata	96 hours
	Acute EC50 0.39 mg/l Fresh water	Crustaceans - Ceriodaphnia dubia - Neonate	48 hours
	Acute EC50 302 µg/l Fresh water	Daphnia - Daphnia magna - Neonate	48 hours
	Acute LC50 1400 µg/l Fresh water	Fish - Pimephales promelas	96 hours
	Chronic NOEC 1 mg/l Fresh water	Algae - Pseudokirchneriella subcapitata	96 hours
	Chronic NOEC 83 µg/l Fresh water	Daphnia - Daphnia magna - Neonate	21 days
1,2-benzisothiazol-3(2H)-one	Acute EC50 0.11 mg/l	Algae - Selenastrum capricornutum	72 hours
	Acute EC50 97 ppb Fresh water	Daphnia - Daphnia magna	48 hours
	Acute LC50 10 to 20 mg/l Fresh water	Crustaceans - Ceriodaphnia dubia	48 hours
	Acute LC50 167 ppb Fresh water	Fish - Oncorhynchus mykiss	96 hours

Persistence and degradability

Bioaccumulative potential

Product/ingredient name	LogP _{ow}	BCF	Potential
1-aminopropan-2-ol	-0.96	0.11	low
Alcohols, C12-15, ethoxylated	2.03 to 6.24	-	high

Mobility in soil

Soil/water partition coefficient (K_{oc})	: Not available.
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Other adverse effects	: No known significant effects or critical hazards.
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Section 12. Ecological information

Section 13. Disposal considerations

Disposal methods : Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Empty containers or liners may retain some product residues. Empty containers retain product residue and can be hazardous. Care should be taken when handling emptied containers that have not been cleaned or rinsed out.

Section 14. Transport information

	DOT Classification	IMDG	IATA
UN number	Not regulated.	Not regulated.	Not regulated.
UN proper shipping name	-	-	-
Transport hazard class(es)	-	-	-
Packing group	-	-	-
Environmental hazards	No.	No.	No.

Additional information

Special precautions for user : **Transport within user's premises:** always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.

Transport in bulk according to IMO instruments : Not available.

Section 15. Regulatory information

U.S. Federal regulations

Clean Water Act (CWA) 311

None of the components are listed.

Clean Water Act (CWA) 307

None of the components are listed.

Clean Air Act Section 112(b) Hazardous Air Pollutants (HAPs)

None of the components are listed.

CERCLA: Hazardous substances.

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302).

Reportable quantity : 1,4-dioxane: 100 lbs. (45.4 kg); ethylene oxide: 10 lbs. (4.54 kg); 2,2'-iminodiethanol: 100 lbs. (45.4 kg); sodium hydroxide: 1000 lbs. (454 kg); acetaldehyde: 1000 lbs. (454 kg);

SARA 302/304

Section 15. Regulatory information

None of the components are listed.

SARA 311/312

Classification : See GHS Classification in section 2 for hazard class information

SARA 313

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

State regulations

Massachusetts : The following components are listed: 1-AMINO-2-PROPANOL
New York : None of the components are listed.
New Jersey : The following components are listed: 2-PROPANOL, 1-AMINO-
Pennsylvania : The following components are listed: 2-PROPANOL, 1-AMINO-
California

California Prop. 65

Ingredient name	Concentration	
Diethanolamine	Trace	Cancer
Ethylene oxide	Trace	Cancer, Developmental, Reproductive female, Reproductive male
Acetaldehyde	Trace	Cancer
1,4-Dioxane	Trace	Cancer

SCAQMD Rule 1144

The sale or distribution in the SCAQM District of California for metal working fluids or direct-contact lubricants is allowed if EITHER the VOC of the product itself OR the VOC of the diluted product at the point of use is less than the following limits: (1) 75 g VOC/L for metal forming, metal removal, metal treating; (2) 50 g VOC/L for metal protection, direct-contact lubricant. The VOC of this product as sold is:

Product as-supplied : 142 g/l ASTM E1868-10

International regulations

Montreal Protocol

Not listed.

Stockholm Convention on Persistent Organic Pollutants

Not listed.

Rotterdam Convention on Prior Informed Consent (PIC)

Not listed.

UNECE Aarhus Protocol on POPs and Heavy Metals

Not listed.

Inventory list

United States : All components are active or exempted.
Canada : All components are listed or exempted.

Section 16. Other information

Date of issue/Date of revision : 1/24/2021

Version : 1

Quaker Houghton Product Stewardship

Key to abbreviations : ATE = Acute Toxicity Estimate
BCF = Bioconcentration Factor
GHS = Globally Harmonized System of Classification and Labelling of Chemicals
IATA = International Air Transport Association
IBC = Intermediate Bulk Container
IMDG = International Maritime Dangerous Goods
LogPow = logarithm of the octanol/water partition coefficient
MARPOL = International Convention for the Prevention of Pollution From Ships, 1973 as modified by the Protocol of 1978. ("Marpol" = marine pollution)
N/A = Not available
SGG = Segregation Group
UN = United Nations
VOC = Volatile Organic Compound

References : **Safety data sheets of raw materials, global regulatory body information, scientific literature, and testing data .**

▀ Indicates information that has changed from previously issued version.

Notice to reader

This product's safety information is provided to assist our customers in assessing compliance with safety/health/environmental regulations. The information contained herein is based on data available to us and is correct to the best of our knowledge, information and belief at the date of its publication. However, no warranty of merchantability, fitness for any use, or any other warranty is expressed or implied regarding the accuracy of this data, the results to be obtained from the use thereof, or the hazards connected with the use of the product. Since the use of this product is within the exclusive control of the user, it is the user's obligation to determine the conditions for safe use of the product. Such conditions should comply with all regulations concerning the product. The company referenced in this Safety Data Sheet assumes no liability for any injury or damage, direct or consequential, resulting from the use of this product unless such injury or damage is attributable to the gross negligence of such company.

TECHNICAL DATA SHEET

HOCUT® 795

HEAVY DUTY MACHINING AND GRINDING FLUID

HOCUT® 795 is compatible with hard water, and is clean-running and biostable assuring long, odor-free sump life. It is suitable for both centralized systems and single-sump machines. HOCUT® 795 provides high corrosion protection without staining and lubricates machine ways and indexing mechanisms.

Applications

HOCUT® 795 is a versatile, high lubricity, soluble oil metal removal fluid formulated to be used across a wide range of heavy duty machining and grinding applications for ferrous and aluminum metals. This includes high and low carbon steels (e.g. 1018), alloyed steels such as 4130 and 4140, cast iron, nodular and gray iron, 300 and 400 series stainless steel, automotive grade cast aluminum alloys including 308, 319, 356, 380, 384, and 390.

Low foaming characteristics make HOCUT® 795 an excellent choice for gun drilling and other high pressure applications.

Recommendation for Use

Machining: 5 to 10%

Grinding: 5%

Benefits

- Extends tool life
- Improves productivity which reduces cycle time
- Provides superior surface finish
- Lower running concentration
- Biostable formulation extends sump life

Health, Safety and Handling

Please consult the Safety Data Sheet (SDS) for information on storage, safe handling and disposal. The conditions or methods of handling, storage, use and disposal of the product are beyond our reasonable control – we assume no liability for any ineffectiveness of the product or any injury or damage, arising out of or in connection with these conditions.

Properties

PROPERTY	TYPICAL VALUE	UNIT
Appearance	Amber liquid	
Typical pH in use	9	
Density	0.95 7.9	g/cc lbs/gal
Refractometer Factor	1	

All reasonable care has been taken to ensure this publication is accurate upon issue. Such information may be affected by changes subsequent to issue. This Technical Data Sheet is to be used solely for this product. Prior to any use, consult the Safety Data Sheet (SDS) for information on hazard risks and product use parameters. All liability and all warranties express or implied are hereby excluded as to product performance results, the accuracy of these data including any warranty of merchantability or fitness for any purpose. 42795000





Product Data

Hysol MB 50

High performance soluble metalworking fluid

Description

Castrol Hysol™ MB 50 is a high performance soluble metal working fluid, which is formulated without chlorine or formaldehyde releasing agents . It contains a unique additive package that works in synergy to enhance machining performance and surface finish, provide excellent product stability, improve bio-resistance properties and lower the overall operations costs.

Hysol MB 50 is suitable for large central systems and single sump machines.

Application

Hysol MB 50 is especially designed for machining of ferrous and aluminium alloys.

	Cast Iron	Low-medium alloyed steel	High alloyed - stainless steel	Aluminium alloys	Magnesium alloys	Yellow metals
Grinding						
Milling, Turning (general machining)						
Drilling						
Reaming, Tapping						
Broaching						

: core application

: recommended application

: suggested application; please consult Castrol representative

Advantages

- Formaldehyde releasing free to comply with health and safety regulations
- Chlorine and nitrite free to fulfill your local legislation, waste treatment and environmental requirements
- Resists fungal and bacterial attack to extend fluid life and reduce maintenance and machine downtime
- Low foam in recommended water conditions
- Good anti-corrosion properties to protect machine tools and parts
- Advanced lubricity compounds ensure improved tool life and part quality
- Excellent wetting properties reduce coolant drag out and deliver clean machine tools and components
- Unique multi-functional product for multi-metal operations results in product consolidation

Typical Characteristics

Name	Method	Units	Value
Concentrate			
Appearance	Visual		Amber
Mineral oil content	Calculated	Wt%	50
Emulsion			
Appearance	Visual		Milky
pH (5%)	DIN 51369 ASTM E70-97		9.7
Refractometer Factor			1.0

Recommended Concentrations

Grinding	5-7%
General Machining	5-8%
Drilling	6-8%
Reaming and tapping	7-10%
Broaching	8-10%

Water range 100-600 ppm CaCO₃

Additional Information

	Boron	Formaldehyde releasing agent	EP-Ester	Amines	Chlorine
Additives		-			-

Hysol MB 50
04 Jun 2019
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